



School of Computing
Department of Artificial Intelligence and Data Science

10214AD701- MAJOR PROJECT

Academic Year: 2025-2026

Major Project - Abstract Form

STUDENT DETAILS:

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TITLE & ABSTRACT:

Title: Integration of AI-Driven Hazard Detection with Navigation Systems for Enhanced Road Safety

Abstract (Min 200 – Max 250 Words in 1 paragraph) - Road safety is a critical concern worldwide, with accidents often caused by unanticipated road hazards such as potholes, debris, fallen trees, and water logging. This project proposes an AI-driven real-time hazard detection system that leverages computer vision and deep learning to identify such hazards from dashcam or mobile camera feeds. The detection module is powered by the YOLOv8 object detection model, trained on publicly available road hazard datasets and self-collected imagery for increased accuracy in local conditions.

Once a hazard is detected, its GPS coordinates are captured and integrated into a navigation system using Google Maps APIs, enabling immediate hazard marking and driver alerts. The system provides both visual cues on the navigation interface and audio warnings to ensure timely driver response, even in high-speed or low-visibility scenarios.

By combining real-time detection, location mapping, and driver notification, the proposed solution not only assists individual drivers but also enables the creation of a crowdsourced hazard database for wider community benefit. This integration aims to proactively reduce accident risks, improve route planning, and enhance overall passenger safety. The system is lightweight, scalable, and adaptable for use in mobile devices, vehicle infotainment systems, and fleet monitoring platforms.

Keywords (Minimum 6) - Example: AI-Driven Hazard Detection, Road Safety, Computer Vision, Deep Learning, YOLOv8, Real-Time Detection, Navigation Systems, Google Maps Integration, Object Detection, Crowdsourced Hazard Mapping, Dashcam Analysis, Pothole Detection, Obstacle Detection, Driver Assistance Systems, Intelligent Transportation Systems (ITS).

APPROVED BY:

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